ALGEBRA 2 · TOPIC 3 · LESSONS 3-1 AND 3-2

Topic 3 Review: Lessons 3-1 and 3-2

ANSWER KEY

Show your work. Answers are in the answer key that comes with this worksheet in Classboard.

A · CLASSIFY

1. Write in standard form, then give the degree and leading coefficient: $9 - x^2 + 4x^3$

$4x^3 - x^2 + 9$; degree 3; leading coefficient 4

2. Name by degree and number of terms: $-6x^4 + x$

Quartic binomial

3. Is $3x^2 + 2/x$ a polynomial? Explain.

No. $2/x = 2x^{-1}$ has a negative exponent.

4. Write a cubic trinomial with a leading coefficient of −2.

Any like $-2x^3 + 5x - 1$ (degree 3, three terms, leading coefficient −2)

B · END BEHAVIOR AND TURNING POINTS

5. Describe the end behavior of $f(x) = -3x^4 + 2x - 1$.

Down on both ends: as $x \to \pm\infty$, $f(x) \to -\infty$

6. Describe the end behavior of $g(x) = 0.5x^5 - x^3$.

As $x \to -\infty$, $g(x) \to -\infty$; as $x \to \infty$, $g(x) \to \infty$

7. What is the most turning points $h(x) = x^6 - 2x^2 + 1$ can have? The most real zeros?

5 turning points; 6 real zeros

8. A polynomial graph rises on the left, falls on the right, and has 2 turning points. Give a possible degree and the sign of its leading coefficient.

Odd degree, 3 or more (for example 3); negative leading coefficient

C · ZEROS AND GRAPHS

9. For $f(x) = (x + 4)(x - 1)(x - 3)$: list the zeros, the y-intercept and the end behavior.

Zeros −4, 1, 3; y-intercept $f(0) = 12$; falls left, rises right

10. Find the average rate of change of $f(x) = x^3 - 2x$ from $x = -1$ to $x = 2$.

$f(-1) = 1, f(2) = 4$; rate $= (4 - 1)/(2 - (-1)) = 1$

11. For $p(x) = x^3 - 9x$: factor it, find the zeros, then use $p(-2)$ and $p(2)$ to say where the relative max and min are.

$x(x-3)(x+3)$; zeros −3, 0, 3; $p(-2) = 10$ so a relative max is between −3 and 0; $p(2) = -10$ so a relative min is between 0 and 3

D · ADD AND SUBTRACT

12. $(2x^3 + 5x^2 - x) + (x^3 - 5x^2 + 7)$

$3x^3 - x + 7$

13. $(6x^2 - 4x + 3) - (2x^2 + 4x - 5)$

$4x^2 - 8x + 8$

14. $(x^4 - 3x^2 + 2) - (x^4 + x^3 - 3x^2)$

$-x^3 + 2$ (the x^4 and x^2 terms cancel)

15. $(4a^2 - 3ab + b^2) - (a^2 + ab - 2b^2)$

$3a^2 - 4ab + 3b^2$

E · MULTIPLY

16. $(3x + 2)(x - 4)$

$3x^2 - 10x - 8$

17. $(x - 1)(x^2 + 3x - 2)$

$x^3 + 2x^2 - 5x + 2$

18. $-2x(x + 5)(x - 1)$

$-2x^3 - 8x^2 + 10x$

19. $(x + 3)^3$

$x^3 + 9x^2 + 27x + 27$

F · APPLY IT

20. A box is made from a 14 cm by 9 cm sheet by cutting an x cm square from each corner. Write $V(x)$ in standard form and give the values of x that make sense.

$V(x) = x(14 - 2x)(9 - 2x) = 4x^3 - 46x^2 + 126x$; $0 < x < 4.5$